

VAST Challenge 2022

Mini-Challenge 1: EngageTown

Solution Document

Team 13 — 大数据 2302

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GitHub: <https://github.com/Kkkkkkkkkkkkenny/VAST-Challenge-2022-Mini-Challenge-1>

目录

1. Introduction
2. Q1: Population Characteristics
 - 2.1 Analysis Approach
 - 2.2 Key Findings
 - 2.3 Deep Insights
3. Q2: Social Activity Patterns
 - 3.1 Analysis Approach
 - 3.2 Ten Social Patterns
 - 3.3 Deep Insights
4. Q3: Economic & Industry Analysis
 - 4.1 Analysis Approach
 - 4.2 Key Findings
 - 4.3 Deep Insights
5. Q4: Community Overview
 - 5.1 Integrated Dashboard
 - 5.2 Comprehensive Insights
6. Summary & Recommendations
7. Technical Methodology
8. References

1. Introduction

This document presents our solution to Mini-Challenge 1 (MC1) of the IEEE VAST Challenge 2022. The challenge asks participants to analyze the EngageTown community dataset — a simulated environment containing 1.14 billion activity logs from 1,011 residents over a 15-month period (March 2022 – May 2023).

Our analysis addresses four core questions: (1) What are the demographic characteristics of EngageTown residents? (2) What notable patterns exist in social activities? (3) What is the dominant industrial basis of EngageTown? (4) Provide a comprehensive overview of the community.

We employ a combination of statistical analysis, social network analysis (SNA), and interactive visualization built with Python, Streamlit, Plotly, and NetworkX. Our approach follows Munzner's (2009) nested model, ensuring analytical rigor at every stage from problem definition to visual encoding.

2. Q1: Population Characteristics of EngageTown

2.1 Analysis Approach

We analyzed 1,011 residents across six demographic dimensions: age, education level, household size, children status, interest group membership, and economic indicators (available balance and joviality index). Nine charts were produced to systematically explore each dimension.

2.2 Key Findings

Finding	Key Data	Chart
Age Structure	Mean 39.1, Median 39, 30-44 peak	Fig 1.1, 1.2
Education	"Some college" most common, then Bachelor's	Fig 1.3
Edu-Balance Link	Graduate > Bachelor > High School > Low	Fig 1.4
Household Size	Mean 2.0, 37% are 2-person households	Fig 1.5
Children	~30% of residents have children	Fig 1.6
Interest Groups	10 groups (A-J), ~101 each, uniform	Fig 1.7
Economy	Balance right-skewed, Joviality ~Normal ($\mu=0.504$)	Fig 1.8, 1.9

2.3 Deep Insights

The demographic structure reveals a self-reinforcing cycle: working-age population supports small businesses, whose employment opportunities attract more young workers. However, the extremely low proportion of elderly residents suggests a lack of intergenerational succession, and the small-family trend means long-term growth depends on external migration.

3. Q2: Social Activity Patterns

3.1 Analysis Approach

We constructed a social network from co-location activity logs and applied Louvain community detection, centrality analysis, and temporal pattern analysis. Twelve charts were produced covering network structure, temporal dynamics, and spatial patterns.

3.2 Ten Social Patterns

Pattern	Finding	Key Metric
1. Near-normal degree dist.	Similar social circle sizes, no super-connectors	Fig 2.1
2. Strong community	28 Louvain communities, $Q=0.5194$	Fig 2.2
3. High local clustering	Avg clustering 0.0714 >> density 0.1738	Fig 2.3
4. Venue type preference	Apt 37.6% > Work 24.2% > Restaurant 21.3%	Fig 2.4
5. Circadian rhythm	Peak 17:00-19:00, weekend +2h shift	Fig 2.5, 2.6
6. Age-connection curve	25-45 most active, declines with age	Fig 2.7
7. Bridge individuals	Top betweenness centrality >> average	Fig 2.8
8. Interaction heterogeneity	Edge weights right-skewed ($\mu=28.3$, $\max=963$)	Fig 2.9
9. Social/leisure dominant	Social+leisure > commute	Fig 2.10
10. Spatial clustering	Venues cluster in 3 social centers	Fig 2.11, 2.12

3.3 Deep Insights

The tension between near-uniform degree distribution (Pattern 1) and strong modular structure (Pattern 2) is notable: social opportunities appear equal, but information flow barriers exist between communities. The fact that leisure spending rivals food spending (Pattern 9) suggests residents treat socializing as a basic need rather than discretionary consumption.

4. Q3: Economic & Industry Analysis

4.1 Analysis Approach

We analyzed 253 employers, their financial flows, wage structures, and building types to characterize EngageTown's economic landscape. Nine charts cover employer size, financial flows, expense composition, wage distribution, and building inventory.

4.2 Key Findings

Finding	Key Data	Chart
Micro-enterprise	All 253 employers: 2-9 employees, mean 5.2	Fig 3.1
Financial structure	Wages = sole income (\$55.6M/yr)	Fig 3.2, 3.3
Wage distribution	Hourly \$10.01-\$40.86, mean \$19.22	Fig 3.4, 3.5
Building inventory	1,042: 526 residential, 512 commercial, 4 schools	Fig 3.6
Industry classification	Only 28/253 (11%) identifiable (F&B)	Fig 3.7, 3.8
Daily cash flow	Biweekly wage cycle, spending follows income	Fig 3.9

4.3 Deep Insights

The highly decentralized structure is itself a source of resilience — no single firm is "too big to fail." However, this comes at the cost of a wage ceiling effect (limited upward mobility for skilled workers) and an innovation dilemma (micro-enterprises lack R&D capacity). The 89% undifferentiated commercial sector suggests a potential classification problem or deliberate data abstraction in the dataset.

5. Q4: Community Overview

5.1 Integrated Dashboard

We synthesized all dimensions — demographics, social networks, economy, and spatial layout — into a comprehensive community profile. Four overview charts provide a holistic view.

Dimension	Key Indicator	Chart
Who We Are	1,011 residents, mean age 39, household size 2.0	Fig 4.1
How We Live	80,483 connections, 28 communities, 12 bars + 20 restaurants	Fig 4.2
Our Economy	253 small businesses, \$55.6M annual wages, housing top expense	Fig 4.3
Our Places	Bars (red), restaurants (green), schools (blue) on map	Fig 4.4

5.2 Comprehensive Insights

EngageTown's core trait is "hidden fragility within apparent balance." Population-economic balance, spatial balance, and social balance coexist, but three structural risks lurk beneath: (1) demographic cliff — aging without replacement, (2) economic monotony — micro-enterprise dependency, (3) social fragility — modular communities with weak inter-group bridges.

6. Summary & Recommendations

Based on our comprehensive analysis, we identify EngageTown as a community characterized by "hidden fragility within small-scale equilibrium." We propose three directions for improvement:

Space Planning: Convert 20% of vacant commercial buildings into innovation incubation zones, offering tax incentives to attract technology-oriented micro-enterprises.

Social Infrastructure: Establish cross-community event programs to bridge modular social groups, strengthening information flow and social cohesion.

Economic Diversification: Develop skill upgrading programs targeting the low-education population to break the wage ceiling and enable economic mobility.

7. Technical Methodology

Our analysis pipeline consists of three stages:

Data Processing: Python + Pandas for cleaning, fastparquet for efficient storage. 14 parquet files (~1.3MB total) derived from ~3GB raw CSV.

Network Analysis: NetworkX for graph construction and analysis. Louvain algorithm for community detection (28 communities, $Q=0.5194$). Degree and betweenness centrality on the largest connected component.

Visualization: Plotly for interactive charts, Streamlit for the dashboard application, Kaleido for static PNG export. Munzner's nested model guides all visual encoding decisions.

The project follows a reproducible pipeline: `process_data.py` → `common.py` → `app.py` (dashboard) and `export_answer_sheet.py` (static HTML answer sheet).

8. References

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